

Design

Encoder: (292k params) I used three 1D-conv blocks (9→64→128→256 channels, kernel 7, stride 2) with a global average pooling to a 256-d vector. My scoring is done global rather than per-window since I want to rescale everything by a shared ruler instead of re-centering each sample on itself. The DC component of the total-acc channels encodes device orientation relative to gravity.

Projector: (1.17 params) I went with Linear 256→960, GELU, Linear 960→960, LayerNorm, then a fixed scale of (mean L2 norm of the frozen embedding table, 3.73) / $\sqrt{960}$ with a learnable scalar gain. This means the sensor embedding is in the same magnitude as real token embeddings, so the frozen network neither swamps nor ignores it.

Context path: The spec prompt is tokenized raw ('add_special_tokens=False', no chat template; SmolLM2-Instruct's BOS is the chat delimiter '<|im_start|>', which does not belong in a raw prompt). Prefix (28 tokens) and suffix ('\nActivity:', 3 tokens) are embedded once through the frozen table, and the projected sensor vector is concatenated between them as 'inputs_embeds' (28 + 1 + 3 tokens). A decode round-trip check asserts the reconstructed prompt matches the spec exactly. The final-position hidden state feeds a trainable 960 → 6 linear head. The LLM ('AutoModel', no LM head allocated) is fully frozen and kept in eval mode, but the forward is not wrapped in 'torch.no_grad()', so gradients reach the encoder/projector through all 32 frozen layers.

Trainable parameters: Direct: 293,190. Context: 1,468,615 (encoder 291,648 + projector 1,171,201 + head 5,766). Budget: 10M

Training setup

I used the standard "42" seed for all runs. Official subject-wise train/test split; validation = subjects {3, 16, 22, 23}, drawn deterministically from the 21 training subjects (never window-level since windows overlap 50%, so random splits would leak). AdamW, weight decay 1e-4, cosine schedule, best-val-macro-F1 checkpointing. Direct: lr 1e-3, 30 epochs, batch 128. Context: lr 3e-4, 2-epoch warmup, 25 epochs, batch 64, fp32 (GTX 1080 Ti). Context training takes ~45 min; the direct model ~2 min.

Condition	Macro-F1
Direct sensor classifier	0.9111
Context-embedding model	0.9037
Context, shuffled embeddings (3 derangements)	0.1661 ± 0.0032
Capacity-matched direct (ablation)	0.9112
Random-init frozen LLM (ablation)	0.9090
Constant-majority floor	0.0514

Sensor-dependence check: Projected test embeddings are permuted with derangements (no example keeps its own embedding), three seeds, no retraining. Performance collapses to 0.166. an exactly

uniform-chance macro-F1 for six classes. Since the prompt is constant, the sensor embedding is the only per-example input, so this collapse was near-guaranteed; it confirms correct wiring and no label leakage, but cannot by itself show the LLM uses the sensor input in an interesting way. The constant-majority floor (0.051) is reported because macro-F1 of a degenerate predictor is well below 1/6.

Interpretation

- **Context = direct (-0.007)** The frozen LLM reads a continuous sensor embedding well enough to match a purpose-built classifier to within one point of macro-F1.
- **Capacity-matched direct = direct (0.9112 vs 0.9111).** The projector's extra 1.2M parameters contribute nothing on their own, so the context model's small deficit is attributable to routing through the LLM.
- **Random-init LLM = pretrained (0.9090 vs 0.9037).** A frozen transformer with the same architecture but random weights matches marginally exceeds the pretrained one. It seems that language pretraining contributes nothing measurable here: the frozen backbone functions as a fixed high-dimensional feature map, and any sufficiently large fixed nonlinearity would do.

Limitations

- One 960 dimension vector per window is a severe bottleneck by design; multiple soft tokens per window were out of scope.
- The linear readout sits on the last hidden state; sensor information could be usable by the LLM yet not linearly decodable there.
- UCI HAR is small and single-modality; conclusions about the shared-interface design need multi-sensor, multi-task evidence.
- Both models' confusion matrices are dominated by sitting to/from standing (~140–150 of ~260 errors), a genuine ambiguity of near-static windows with similar orientation; all conditions classify laying essentially perfectly (shocking).

My Recommendation

This approach is validated, however, I have a couple of thoughts regarding whether the model has earned its place:

1. We have confirmed that a sensor can be attached to a frozen LM through on continuous embedding. A scale-matched embedding routed through 32 frozen layers trains stably and reaches within 0.007 macro-F1 of a purpose-built classifier, and the shuffle check confirms full sensor dependence. This is a strong development supporting the model's existence.
2. However, I fail to see how the language model added value for this task. The pretrained backbone performs identically to a random frozen one, and both roughly match the direct classifier. Closed-set single-sensor classification with a constant prompt gives the LM no opportunity to use what pretraining provides; it is reduced to an expensive fixed feature map.

Therefore, my recommendation is to not develop this path further for activity classification alone. Instead, shoot for where our results justifies one more iteration, aimed at where language priors could actually bind: a varying prompt, multiple sensor modalities sharing one interface, or open-vocabulary activity descriptions. Those are the cheapest experiments that could show the pretrained LM doing something a random feature map cannot.